

SHETH MVLU COLLEGE

SUBJECT:- WSN

AIM:

Mote-to-Mote Radio Communication Simulation

Simulate packet transmission between two motes and analyze signal strength and packet loss under varying radio gain values.

Code:

Mote.ned:

```
package motecommunication;
```

```
simple Mote
```

```
{  
  parameters:  
    double sendInterval @unit(s);  
  gates:  
    input in;  
    output out;  
}
```

RadioNetwork.ned:

```
package motecommunication;
```

```
network RadioNetwork
```

```
{  
  submodules:  
    mote1: Mote  
    {  
      parameters:  
        sendInterval = 1s;  
    }  
  
    mote2: Mote  
    {  
      parameters:  
        sendInterval = 0s;  
    }  
  
  connections:  
    mote1.out --> mote2.in;  
    mote2.out --> mote1.in;  
}
```

SHETH MVLU COLLEGE

SUBJECT:- WSN

Mote.cc:

```
#include <omnetpp.h>
```

```
class Mote : public cSimpleModule
```

```
{
```

```
private:
```

```
    cMessage *timer;
```

```
    int received;
```

```
protected:
```

```
virtual void initialize()
```

```
{
```

```
    received = 0;
```

```
    if(strcmp(getName(),"mote1")==0)
```

```
    {
```

```
        timer = new cMessage("sendTimer");
```

```
        scheduleAt(simTime()+1,timer);
```

```
    }
```

```
    else
```

```
        timer = NULL;
```

```
}
```

```
virtual void handleMessage(cMessage *msg)
```

```
{
```

```
    if(msg==timer)
```

```
    {
```

```
        EV << "Sending Packet\n";
```

```
        cMessage *pkt=new cMessage("DataPacket");
```

```
        send(pkt,"out");
```

```
        scheduleAt(simTime()+1,timer);
```

```
    }
```

```
    else
```

```
    {
```

```
        received++;
```

```
        EV << "Packet Received Count="
```

```
        << received << "\n";
```

```
        delete msg;
```

```
    }
```

```
}
```

```
virtual ~Mote()
```

```
{
```

SHETH MVLU COLLEGE

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```
cancelAndDelete(timer);  
}  
};  
Define_Module(Mote);
```

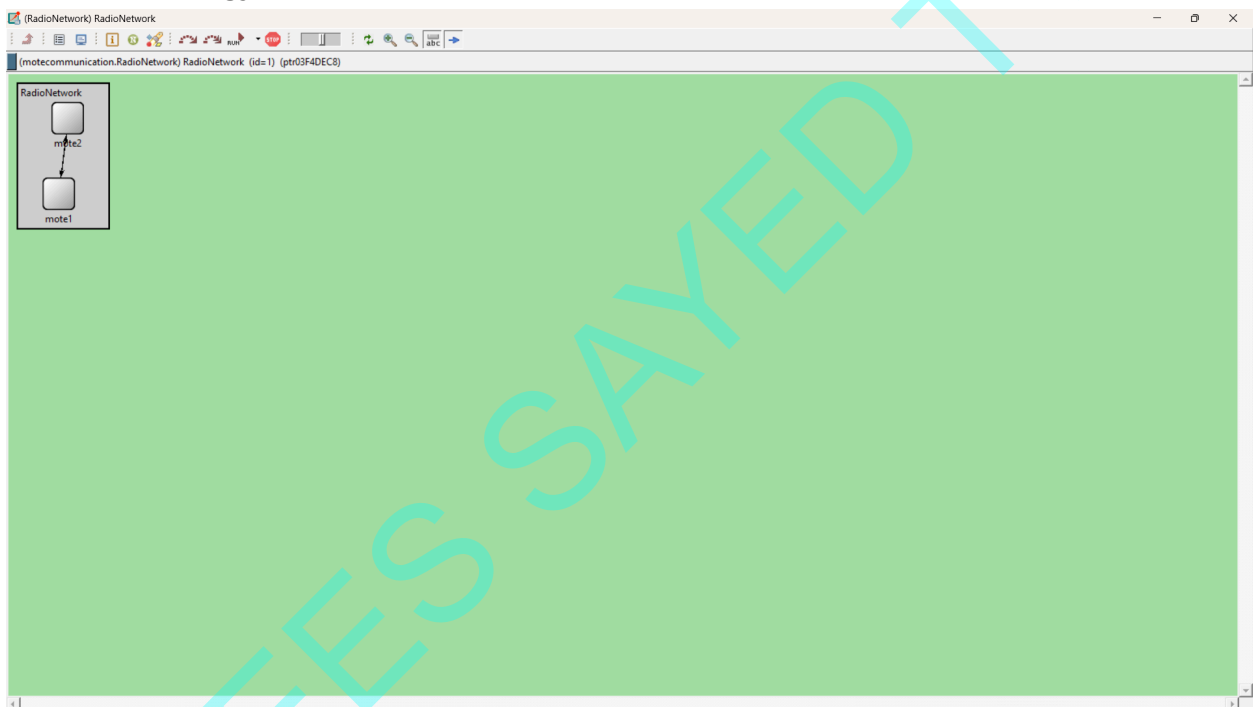
omnetpp.ini:

```
[General]  
network = motecommunication.RadioNetwork
```

```
sim-time-limit = 20s
```

OUTPUT:

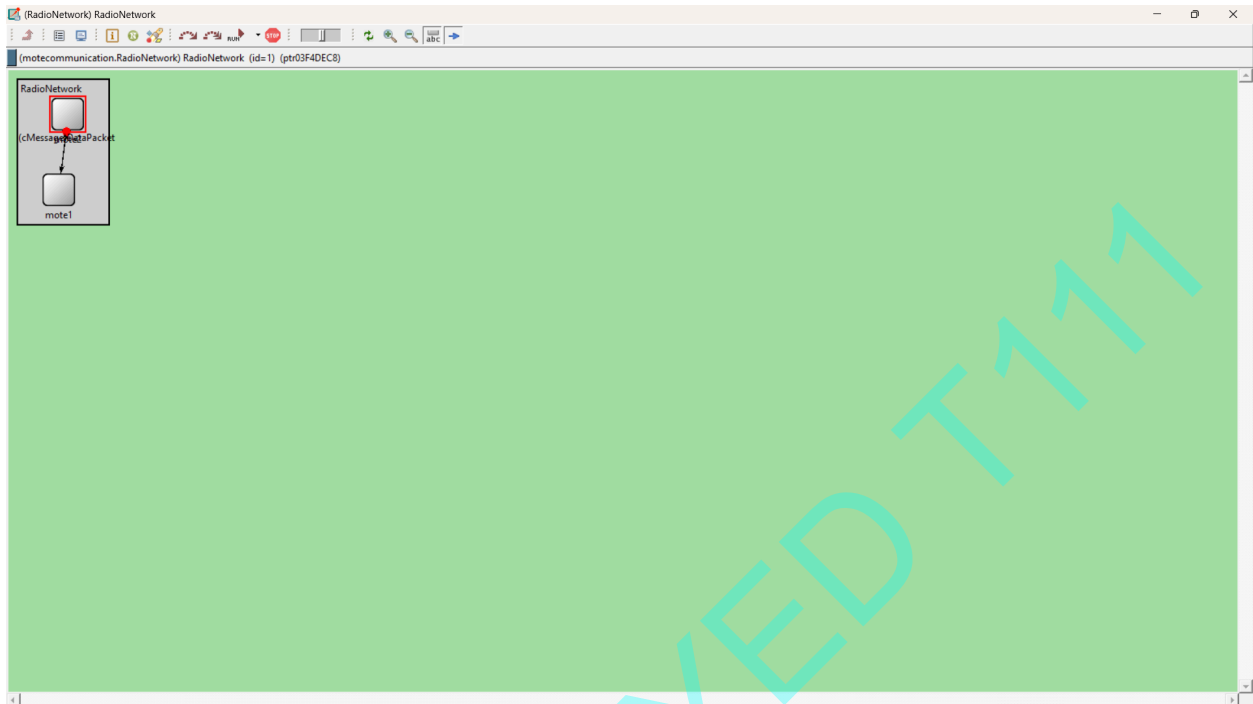
Network Topology (mote1 ↔ mote2)



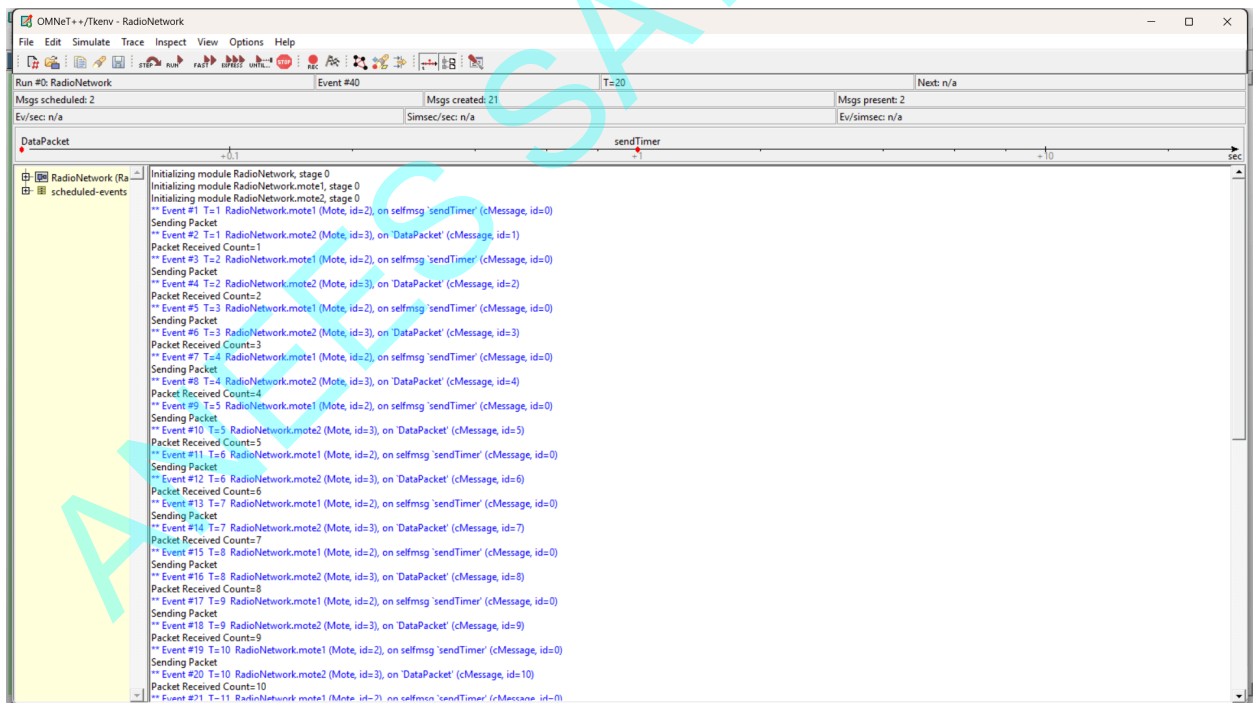
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Simulation Run:



Event Log Showing Packet Transmission:



Result

The packet transmission between two motes was successfully simulated. Packets sent by **mote1** were received by **mote2**, demonstrating mote-to-mote communication in the network.